

Final Project

Motivation:

A significant statistical association between a SNP and a phenotype of interest is not enough to conclude that the SNP is *causing* that phenotype. The association could arise indirectly, through confounding or linkage, rather than a direct causal effect. To distinguish these possibilities, we need to conduct mediation tests. We would start by running a QTL analysis with the phenotypic data against genotype information, measured by markers across the genome. Complex traits are often not affected by DNA variants directly, but through their effect on gene expression. So the next step is to use gene expression as an intermediate phenotype: we find eQTLs, associating the same genotype markers with gene expression levels. Lastly, having both a QTL and an eQTL for a given locus, you are going to test whether the SNP's effect on the phenotype is genuinely mediated through gene expression, i.e., test for causality.

General guidelines:

- This assignment should be submitted in pairs, as in HW2.
- **Start by reading the whole assignment before you begin with implementation, so that you can plan your code accordingly and won't waste time on modifying it along the way.**
- Some parts of this assignment (mostly section 5) can be computationally heavy. If your running times are long, try to run the code on a server such as Google Collab. You can also try to implement the script in a NumPy-vectorized fashion to speed up execution.

Submission guidelines:

- Submit a single zip file named **Project_<your_id1>_< your_id2>.zip**
- The zip file should contain:
 - Your code as an .ipynb file (the Jupyter notebook/Collab format).
 - A report, as a PDF file.
- Notes on your code:
 - All the code blocks should be executed. You do it by simply executing your code and saving it as an ipynb file. Don't save it as a PDF with the outputs.
 - Submit a clean, documented, and organized code. In case of wrong answers, it will assist us in finding the cause and reduce point deduction.
 - Don't include the answers to the theoretical questions in the code. These should be written in your report!
- Report expectations:
 - The report should be an organized, professional file, summarizing the experiment you did. We review and grade your work (mostly) by reading the report. Therefore, it should contain all the answers, results, discussion, plots, etc. Don't leave them in the code.
 - Be mindful when using generative AI for the report. It is okay to use it for sharpening your writing, but we don't want to read whole pages generated by it. Please notice that they tend to output long, redundant explanations. Find the sweet spot between too much and too little.

Section 1: Define research objective

We would start by choosing the phenotype and gene expression data we're working with.

1. **Choose two phenotypes of interest.** You can take one from HW2.

Notice that some phenotypes are results of the same experiment over different time intervals.

For example, phenotypes 1219, 1222, 1223, 1224, are all records of the same collection¹:

locomotion in cm after morphine injection. 1219 measures minutes 0-15, 1222 measures minutes 15-30, and so on. Your phenotypes of choice should be of different collections, i.e., you cannot choose 1219 and 1222.

The phenotypes don't have to be from the same paper.

In your Report: Present the phenotypes you are working with. How do the researchers measure them? What are the experimental settings?

2. **Choose two of the following gene expression datasets.**

Your choice of gene expression reference should be related to your phenotypes. For example, if you look at locomotion after morphine injection, use gene expression from both brain and liver tissues as relevant intermediate traits, as the brain is known to be influenced by psychoactive drugs, and the liver is responsible for the metabolism of drugs.

Download from [GEO database](#), or use the Python command: `GEOparse.get_GEO(Access_ID, output_path)`.

Make sure you download the normalized data.

In your Report: Explain why the chosen datasets are relevant to your target phenotypes.

Tissue / Cell type	Access ID
HSC	GDS1077
Myeloid	GSE18067 ²
Blood stem cell	GSE18067
Erythroid	GSE18067
Hypothalamus	GSE36674
Kidney	GSE8356
Liver	GSE17522

¹ Some of these collections have an entry that summarizes the whole experiment, like phenotype 1231, which tracks the locomotion in cm after morphine injection, in minutes 0-180. Working with such phenotypes will make it easier to interpret the results but may lead to a reduction in signal.

² Notice that Myeloid, Blood stem cell, Erythroid are all GSE18067. this is one GEO series containing multiple cell types that need to be subset out.

Section 2: QTL analysis

In this section, we will find SNPs with a statistically significant association to your phenotypes³, just like in HW2.

1. Copy the QTL analysis pipeline from HW2 to your new script, and make the following adjustments:
 - a. We don't need all the functions! **We only need the functions that parse the Excel files, take a phenotype ID, run an F-test on your selected phenotype using each of the SNPs in the genotyping file, and output a P-value per SNP.** You can remove the ones related to ANOVA, Manhattan plot, etc.
 - b. In the F-test, use the linear regression model in which heterozygous markers are **considered**. Encode B = 0, H = 1, D = 2. It is okay if a SNP doesn't have heterozygosity in any strain.
 - c. Use BH-FDR correction for multiple testing, and not Bonferroni. Record the q-values for each SNP, rather than correcting the significance threshold α itself (as you would for Bonferroni).
 - d. The pipeline's output should be a CSV of the significant SNPs, with the Locus name ('rs13475979') and adjusted P-value. If there are no significant SNPs, output the 15 SNPs with the lowest P-value.
2. Run the pipeline on the two phenotypes you chose.
In your Report: Add your results. How many significant QTLs did you find for each phenotype?

Section 3: eQTL analysis

In this section, you will investigate the genetic basis of transcriptional responses of two tissue/cell types of your choice to different stimulations in BXD recombinant inbred lines.

1. **Apply the following preprocessing steps to the two gene expression datasets:**
 - a) Download the gene expression data of the two tissue types you chose.
 - b) Merge the data file with the annotation file to get your input matrix. In your input matrix, rows should be annotated with gene identifiers (either gene symbol or Entrez IDs); columns are BXD strain names.
 - c) Remove rows with no gene identifier.
 - d) Remove rows with low maximal value (choose a threshold).
 - e) Remove rows with low variance (choose a threshold).
 - f) The data may contain multiple rows (probes) for the same gene. To have a single row per gene, either select one probe while removing the others (e.g., select the highest- variance probe) or calculate their average.
 - g) Use Genotyping file from assignment 2; use strains that appear both in the genotype file and in the expression data. Use only representative genomic loci by filtering those loci that have the same information as neighboring loci across the BXD strains under study.
 - h) Gene expression data should be averaged across different individuals of the same strain before applying the association test.

In your Report: Describe the data preprocessing steps with details. If you had any problems with parsing, explain how you solved it. Mention what were the thresholds you used in steps d & e, and how you chose them.

³ The downstream analysis would be way more interesting if your phenotype of choice has significant QTLs. It would be better to test a few phenotypes in this QTL pipeline before you commit to two.

2. **Run an association test** on all genes, using each of your selected representative SNPs in the genotyping file, as we defined in class. Heterozygous markers should be considered.

For each gene, select those eQTLs that are best associated with the expression of the gene (genes with weak associations for all SNPs should be excluded).

Account for multiple testing with BH-FDR correction; don't use permutations here.

3. **Results Analysis:**

- a) How many different significant eQTLs did you find? How many of them are *cis*-acting⁴ and how many are *trans*-acting?
- b) What is the distribution of the number of genes associated with a given eQTL?
- c) Plot the number of genes associated with each eQTL across the genome.
- d) Compare the distribution of P-value scans between *cis*-associated genes and *trans*-associated genes.
- e) Create a scatter plot visualization (as seen in lecture) and indicate *trans*-acting hotspots and clarity of *cis*-acting eQTLs within it.

In your Report: Describe how you chose the final collection of eQTLs. Then, answer questions a, b, c, d, e for tissue 1 and tissue 2.

Section 4: Combining QTL and eQTL results

Here, we compare the results of Section 2 and Section 3.

1. Were any SNPs found to be both QTL and eQTL? Generate a Venn diagram showing the distribution (separated into QTL, eQTL of tissue one, eQTL of tissue 2).
2. Does gene expression data provide any added value for the identification of QTLs?
3. From your experience, what conclusion can be drawn about limiting GWAS to those DNA variants that are associated with at least one expression trait? Positive / Negative sides?

In your Report: Add the answers to all questions + the diagram.

Section 5: Causality analysis

Here you are going to apply a causality test on the results from the previous parts. Follow the model guidelines presented in class.

1. Run a causality test on each pair of gene and phenotype where both the QTL and the eQTL are in a nearby genomic position (or, of course, have the same position). You need to implement the mediation test functions yourself; Do not use a ready-made Bayesian network library.
 - a) Explain the causality test model we use in detail.
 - b) Report the predicted relations among the QTL, associated gene, and phenotype.
2. Apply a permutation test to get statistical significance:
 - a) Explain the choice you made in the design of the permutation test in detail.
 - b) Run the test on 10 triplets⁵. Explain how you chose them and share the results.

In your Report: answer the questions above and elaborate on the decisions you made throughout the way.

Then, discuss the results: do they align with your expectations? Is there anything that could have been done to improve the experiment results? What were the problems/caveats in our experimental settings? Feel free to include additional experiments or visualizations if necessary.

⁴ For this assignment, we consider eQTLs to be *cis*-acting when the distance between the SNP and the gene is 2Mbp or less.

⁵ We requested 10 triplets because execution times can be long. If runtime isn't actually a bottleneck in your implementation, you can run the test on more / all triplets.